Shell & Kernel Programming
Simple Filters

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Filters are commands which can accept input from the standard input manipulate the stream and produce the output at the standard output. Filters are standard tools in the arsenal of the UNIX Shell programmer and are often used in conjunction with each other. Some of the commonly used filters in UNIX are listed in the Table 1

Table 1: List of commonly used UNIX Filters

Filter	**Effect**
`more`	more is used to paginate long scrolling text outputs.
`wc`	wc can be used to count lines, words and characters.
`cmp`	compare two files to find the differing bytes and their locations.
`diff`	diff is used to print a routine that allows one file to be converted to another
`comm`	comm is used to find the common content in two sorted files.
`head`	head displays lines from the top of a file.
`tail`	tail displays lines from the end of a file.
`cut`	cut is used to vertically split a file.
`paste`	paste is used to vertically join two files.
`sort`	sort is used to sort the contents of a file.
`tr`	tr converts a set of characters to another set of characters.
`uniq`	uniqu (note the missing e) returns unique lines in a file.
`nl`	add line numbers to a file for the purpose of display.
`grep and egrep`	the goto tool for in text searching in UNIX.

As is the case with almost all commands in UNIX all the filters listed above take a large number of options which fine tune their outputs.

0.1 grep and egrep

`grep` is an extremely powerful filter to search content in files. The primary usage of `grep` is extremely simple but grep also accepts quite a few options. Some of the more commonly used options are listed in Table 2

The power of grep is further enhanced by the use of regular expressions. The grep regular expressions are heavily inspired by the PERL language. Some of the common regular expressions are shown in Table

0.2 Assignments

1. Create a variable storing the number of words in the text file data.txt.

Table 2: List of common options in grep

Options	Effect
-n	show line numbers.
-l	show just the filenames.
-v	show lines not containing the search text.
-c	count the number of occurrences of the search text.
-i	perform case ignorant search.
-w	searches for whole words only.
-n	perform numeric sort.

Table 3: List of regular expressions used in grep

Expression	Effect
.	A single character.
*	zero or more occurrences of the last character.
[pqr]	one single character from among p, q and r.
[p̂qr]	one single character not among p, q and r.
[a-z]	one single character in the specified range.
^x	line starts with x.
x$	line ends with x.
\{m\}	the last character repeated m times.
\{m,n\}	the last character repeated a minimum of m times and a maximum on n times.
+	the last character repeated one or more times. (egrep only).
?	the last character repeated zero or one times. (egrep only).
x|y	either x or y.
(xyz)|(abc)	either of the two sequences.

2. Create two text files containing just the designation and salaries in the file data.txt.

3. Create a file containing just the filenames and modification times for the files in your current folder.

4. Sort the files in your current folder only on the number of links that the files contain.

5. Sort the data in the file data.txt only on the year of birth.

6. How many persons are there in each designation in the file data.txt?

7. Find all sequences of digits containing either 5 or 6 consecutive digits in the file data.txt?
8. Find in how many lines does accounts appear in the file data.txt?
9. Find all occurrences of True, trueman, truman, Truman in the file.
10. How many persons have their employee id start with a 1?
11. How many directories exis in your current folder?
12. Who is the person whose salary does not end with a 0?
13. Who gets the highest salary in the file data.txt?